

**TED(10)- 5045
(REVISION-2010)**

**MODEL QUESTION PAPER (REV-10)
FIFTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY
(MD only)
TECHNOLOGY AND MEDICAL SCIENCE**

(Maximum Marks: 100)

TIME: 3HRS

PART - A

Answer all Questions in one or two sentences. Each question carries **2 marks**.

I.

1. Define genetic-engineering?
2. Give an account on enzymes?
3. Give the concept of AI?
4. Define HIS?
5. Define nuclear magnetic resonance?

5x2=10 marks

PART - B

Answer any **FIVE** of the following questions. Each question carries **6 marks**.

II.

1. State the difference between DNA and mRNA?
2. Give an account on cloning?
3. Explain the basics of protein data bank?
4. State the automated methods in hematology?
5. Explain the applications of MYCIN and EMYCIN?
6. Design the concept of laboratory computerization system?
7. Explain the principle of electron microscope?

5x6=30 marks

PART - C

Answer one full question from each Unit. Each question carries 15 marks.

UNIT I

- III** a) Describe the application of genetic engineering? (5)
(b) Explain the tools for genetic engineering ? (10)

OR

- IV.** a) Explain the mechanism of DNA replication? (8)
b) Explain Characteristics of gene mutation? (7)

UNIT II

- V.** a) State different types of data types? (10)
b) Account briefly application of computer aided drug design? (5)

OR

VI a). Discuss the application and discipline techniques used in AI? (10)

b) Explain logical reasoning and decision making in AI?
(5)

UNIT III

VII a). Explain the basic concept of hospital information system? (8)

b) Describe the basic principle of automated scanning in cervical cancer?
(7)

OR

VIII a) Explain the clinical information system? (8)

b) Give an account on intelligent laboratory system?
(7)

UNIT IV

IX. Explain the principles of scanning and transmission electron microscope with neat sketch?
(15)

OR

X (a) Describe the basic principle of mass spectrometer? (8)

b) Explain NMR spectroscopy? (7)

15x4=60